Fall soil balance

Organic greenhouse tomatoes, in-ground beds. Run right after pull-out so amendments react over winter. All inputs organic-compliant.

Flow: slurry EC → broadfork → leach? → sample → pH → amend.

Step 1 — Slurry EC

`scouting/slurry-ec-test.md` per greenhouse. Record. That's the leach read.

No EC meter → SME test and leach on SME EC > 4.5 dS/m.

Step 2 — Broadfork

Every bed, full pass, tines to depth. Fractures the top ~12 in where greenhouse compaction lives so leach water moves down, not sideways. Replaces the percolation test.

- Insert every 6–8 in, walking backward; one step, one insertion
- Two passes side by side on a 5 ft bed
- Lean back until the soil lifts and cracks; don't flip
- ~15–20 min per 100 ft bed

Step 3 — Leach decision

Read the slurry EC from Step 1 against the salt-buildup band on the slurry card.

Read	Do
Below salt buildup	Skip to Step 4
Salt buildup	Leach
Salt buildup and last M3 Na > 50 ppm, or softened / brackish water	Gypsum first, then leach. Ca displaces Na off exchange sites

- Salty water (softener, brackish well) can't leach. Use rain through vents, rain barrel, trucked water.
- Run the Leach cycle.
- 2 days after the last pulse: slurry EC again (SME Test if no EC meter)
- In band → done. Still salt buildup → another Leach cycle.

One cycle halves salts; that's not always enough.

Reading didn't drop ~40–50% → water isn't moving through. Fix drainage or drip coverage before more water.

Leach Cycle - one size fits all, any soil, any tape - **Run 4 h per day × 7 days** on a daily timer. - Aim for the best coverage: - Wobbler 10ft appart - 4 Drip tapes per bed

Step 4 — Fall soil sample

After the last leach cycle, or right after broadfork if no leach. Leaching strips K, Mg, B; sampling after it makes the Step 6 doses right.

- Before first irrigation of the day
- Wash trowel and plastic bucket first
- 12–15 cores per greenhouse
- Dug knuckle deep, between emitters, skip bed edges
- Mix in and submit ~1lb

Lab order: - Mehlich-3 test - Buffer pH add-on

Step 5 — pH correction

Aim **6.2**. Leaves a season of upward drift before 6.5.

Max 0.5 unit correction per fall; next fall's test finishes the job.

Doses per 100 ft × 5 ft bed, per 0.1 unit. Incorporate. 3–6 months to react.

pH > 6.5 → Tiger 90CR

Sand	Loam	Clay
0.5 lb	0.9 lb	1.8 lb

pH < 6.0 → pelletized calcitic lime.

Sand	Loam	Clay
2 lb	4 lb	8 lb

Flag: well water drifts pH up +0.1–0.3/year. Annual maintenance. pH still up two falls running despite sulfur → test water alkalinity (as $CaCO_3$). Above 200 ppm → acidify fertigation to pH 5.5–5.8 with citric acid.

Step 6 — Fall amendments (immobile nutrients)

Nutrient	Target (M3)	When low
P	75 ppm	Bone meal
K	190 ppm, ≥ Ca ÷ 8, ≥ Mg ÷ 1.4	Sulfate of potash
Ca	800 ppm	Gypsum
Mg	125 ppm, ≥ Ca ÷ 11	Epsom salt
Zn	4 ppm	Zinc sulfate
Mn	25 ppm	Manganese sulfate
Cu	1.5 ppm	Copper sulfate
Fe	60 ppm	Iron sulfate
B	0.75 ppm	Borax / Solubor, tiny dose; toxicity is 2× sufficiency

pH lockup: dose anyway, same fall as sulfur.

Don't overdo it: There are NO proven yield gain associated with nutrients over target while yield losses DEFINITELY HAPPEN! Also, undershoot is fixed with routine fertilizer top off while overshoot on Cu, Mn, B, Mg takes years to fix.

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increase needed = target ppm − current ppm
lb/ft² ≈ increase needed × 0.046 ÷ label % ÷ oxide factor ÷ 1000
lb/bed = lb/ft² × bed top width (center to center without alley) × bed length
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Label % as a decimal: 50% K_2O → 0.50, not 50.

Nutrient	Oxide factor
P	0.44 (label is P_2O_5)
K	0.83 (label is K_2O)
All others	1

Example: K at 100 ppm, target 190. Sulfate of potash, label 50% K_2O → 0.50, oxide factor 0.83.

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lb/ft² = 90 × 0.046 ÷ 0.50 ÷ 0.83 ÷ 1000 = 0.010
bed top 5 ft × 100 ft = 500 ft² → 5 lb
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Incorporate knuckle deep

What not to do

No compost

Blind compost brings salts, P overload, K you didn't budget; it undoes Steps 3 and 6. Skipping it costs nothing for years: OM drops ~0.1–0.2%/year, a 5% soil is 4% in a decade, fertigation supplies the nutrients. Worst case is more frequent irrigation and a yearly broadfork.

Save costs and labor. Skip it.

OM% under 3% : Consider compost carefully for structure, water-holding and the biology that runs the sulfur reaction. Don't do it blindly or you'll ruin the leach and amendment steps !

No cover crop, no tarp

Bare soil in a greenhouse: no rain, no erosion, no leaching, trivial weeds. One winter of cover gains little. Bare and broadforked is fine.

Save costs and labor. Skip it.

Grower deliverables

1. One-page sampling SOP (laminated, header house)
2. Interpretation calculator: M3 in → amendment shopping list out
3. Decision flowchart (the flow at top)

Delegate

Greenhouse manager / fertility person. Sampling can be staff.